

Lab 5

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Deadline: 2022-11-04 22:30:00 Beijing Time

How to submit?

Email: YJ40202@163.com

Subject: [exercise](#)

Attachment: Your java source files.

Note: Don't use package.

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Exercise05_11.java

(Find numbers divisible by 4 or 5, but not both) Write a program that displays all the numbers from 100 to 200, ten per line, that are divisible by 4 or 5, but not both. Numbers are separated by exactly one space.

```
104 105 108 110 112 115 116 124 125 128  
130 132 135 136 144 145 148 150 152 155  
156 164 165 168 170 172 175 176 184 185  
188 190 192 195 196
```

Exercise05_14.java

(Compute the greatest common divisor) A method to find the greatest common divisor of two integers n_1 and n_2 is as follows: First find d to be the minimum of n_1 and n_2 , then check whether d , $d-1$, $d-2$, ..., 2, or 1 is a divisor for both n_1 and n_2 in this order. The first such common divisor is the greatest common divisor for n_1 and n_2 . Write a program that prompts the user to enter two positive integers and displays the gcd.

Here is a sample run:

```
Enter the first number: 6 [Enter]
Enter the second number: 9 [Enter]
GCD of 6 and 9 is 3
```

Exercise05_15.java

(Display the ASCII character table) Write a program that prints the characters in the ASCII character table from ! to ~. Display 10 characters per line. Characters are separated by exactly one space.

```
! " # $ % & ' ( ) *  
+ , - . / 0 1 2 3 4  
5 6 7 8 9 : ; < = >  
? @ A B C D E F G H  
I J K L M N O P Q R  
S T U V W X Y Z [ \  
] ^ _ ` a b c d e f  
g h i j k l m n o p  
q r s t u v w x y z  
{ | }
```

Exercise05_17.java

(Display pyramid) Write a program that prompts the user to enter an integer from 1 to 15 and displays a pyramid, as shown in the following sample run:

```
Enter the number of lines: 8 [Enter]
```

```

      1
    2 1 2
  3 2 1 2 3
4 3 2 1 2 3 4
5 4 3 2 1 2 3 4 5
6 5 4 3 2 1 2 3 4 5 6
7 6 5 4 3 2 1 2 3 4 5 6 7
8 7 6 5 4 3 2 1 2 3 4 5 6 7 8
```

Exercise05_25.java

(Compute π) You can approximate π by using the following summation:

$$\pi = 4 \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \dots + \frac{(-1)^{i+1}}{2i-1} \right)$$

Write a program that displays the π value for $i = 10000, 20000, \dots$, and 100000 .

```
i = 10000 The PI is 3.1414926535900345  
i = 20000 The PI is 3.1415426535898248  
...  
i = 90000 The PI is 3.1415815424786238  
i = 100000 The PI is 3.1415826535897198
```


Exercise05_33.java

(Perfect number) A positive integer is called a perfect number if it is equal to the sum of all of its positive divisors, excluding itself. For example, 6 is the first perfect number because $6 = 3 + 2 + 1$. The next is $28 = 14 + 7 + 4 + 2 + 1$. Write a program to determine whether the given number is a perfect number.

```
Please enter a number:28 [Enter]  
28 is a perfect number!
```

```
Please enter a number:100 [Enter]  
100 is not a perfect number!
```

Exercise05_45.java

Write a program that prompts the user to enter 10 numbers and displays the mean and standard deviations of these numbers using the following formula:

$$\text{mean} = \frac{\sum_{i=1}^n x_i}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n} \quad \text{deviation} = \sqrt{\frac{\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n}}{n - 1}}$$

Enter 10 numbers: 1 2 3 4.5 5.6 6 7 8 9 10 [Enter]

The mean is 5.61

The standard deviation is 2.997943739743404

Exercise05_46.java

(Reverse a string) Write a program that prompts the user to enter a string and displays the string in reverse order.

```
Enter a string: hello [Enter]  
The reversed string is olleh
```

Exercise05_48.java

(Process string) Write a program that prompts the user to enter a string and displays the characters at even positions.

We assume that the index of the first character is one. Here is a sample run:

```
Enter a string: hello, how are you? [Enter]  
e1,hwaeyu
```